

# The Scale Ladder

Building blocks and fields across all layers of matter —  
a pedagogical sketch on inter-scale couplings in  
FFGFT

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## Preface

This document is a *plausibility sketch*, not a derivation. It contains hypotheses, not proofs. Where a quantitative statement is made, it is to be read as an order-of-magnitude estimate, not as a prediction. The central question — *how strongly do the different scales of matter couple geometrically with one another?* — is an open research question, not a solved problem.

The purpose of this document is not to deliver an answer, but to *state the question cleanly* — so that anyone working through FFGFT does not overlook the scale ladder between elementary particles and cosmology as a gap, but recognizes it as research terrain.

FFGFT has so far been worked out *at the lowest layer*: elementary particles, modes on the 4D-torus, winding numbers. The upper boundary — the cosmological maximum scale — is treated separately in Doc. 182. What lies between — atoms, molecules, crystals, gases, liquids, planets, stars, galaxies — is addressed

at scattered points in the document series (Doc. 167 for crystals, Doc. 168 for atoms), but not developed narratively as a coherent scale ladder. That is what this document tries to make up, in the pedagogical style of Doc. 241.

## 1 A generally unsolved problem

Before FFGFT enters the picture, it is worth taking stock of what standard physics actually says about this question — or does not say.

The Standard Model (SM) of particle physics is extraordinarily precise on its own scale. It describes quarks, leptons, and their interactions with an accuracy no other physical theory matches. But it has *no hierarchy theory*: there is no derivation within the SM of why quarks are bound in hadrons, hadrons in nuclei, nuclei in atoms — or why these structures look exactly as they do. That is an observation, not an explanation.

### What the Standard Model has

Standard physics meets this problem through *effective theories*: QCD for quarks and gluons; nuclear physics as a phenomenological extension; atomic physics via QED. The transitions between these levels are not derived — they are empirically adjusted. No one has calculated from the QCD Lagrangian alone why a proton has the mass it has, let alone why a hydrogen atom is built exactly as it is. Lattice field theory can now simulate the proton numerically; *deriving* it — understanding why exactly this structure and no other — remains out of reach.

The closest relative to a cross-scale explanation is the *renormalization group* (RG). The RG describes how coupling constants scale with energy. Asymptotic freedom in QCD is precisely that: at high energy quarks are quasi-free, at low energy strongly bound. That

is a genuine scaling effect. But the RG runs *vertically* through energy scales and says nothing about how the structure at one scale *shapes* the next. It describes the flow of couplings, not the emergence of structures.

## The actual gap

The real question is: why does an atom, as a consequence of the quark structure of the proton, carry exactly the structure it carries? The SM cannot answer this. How the properties of the proton — mass, spin, magnetic moment — emerge from QCD is still numerically hard and conceptually open. And one level up: why do protons and neutrons form exactly the nuclei they form, with exactly these binding energies? Nuclear physics is empirical — not derived from QCD.

This is not a failure of diligence. It is a structural feature of standard physics: it describes the mechanisms *within* each level very well, and the *local transitions* from one level to the next through appropriately chosen coupling parameters. What it does not describe is whether and how a geometry at one scale *propagates through* the scale ladder — whether what is fundamental at the quark level leaves any trace in the structure of the atom or the crystal.

## An empirical argument: external influence as structural proof

There is an argument that requires no theory and yet points directly to the heart of the question.

We influence molecular structures through external fields: an electromagnetic field changes the conformation of a protein; a mechanical pressure shifts a crystallisation pathway; a change in temperature drives a chemical reaction in a different direction. This

influence is not a special effect — it is experimental routine in chemistry, biophysics, and materials science.

But it has a structural consequence that is easily overlooked: *for an external field at a higher scale to be able to alter a molecular structure at a lower scale, coupling channels between these scales must physically exist.* These channels are not a property of the external field. They are a property of the structure itself.

And here comes the decisive step: coupling channels that physically exist are bidirectional. If a macroscopic field configuration can act on the electron shell of a molecule, then conversely the substructure of the molecule — the arrangement of atoms, the electron density, the nuclear geometry — acts on the superordinate structure. Not through external intervention, but as system-internal propagation.

This is not a speculative conclusion. It already shows itself in protein folding: a protein folds into its equilibrium structure without external intervention. The information about *how* it folds resides in the amino-acid sequence — at the chemical scale. This information propagates upward to the macromolecular geometry. That is inter-scale coupling, system-internal, without an external source.

Standard physics describes these channels locally and correctly: Coulomb interaction, van-der-Waals, hydrogen bonds. What it does not describe is whether these local channels are the expression of a *continuous geometric structure* — or merely a collection of accidentally fitting mechanisms. FFGFT claims the former. The SM is silent on the question.

## **Stability as a hidden presupposition**

Behind the SM's silence on inter-scale coupling lies a methodological decision that is rarely made explicit: the SM treats stable superordinate structures as *boundary conditions*, not as *explana-*

*tory targets*. An atom exists — fine, we calculate with it. A crystal exists — fine, we describe its band structure. Why these structures are stable at all, and why they have exactly this form and not another, is not asked. Stability is taken as given.

This conceals a fundamental assumption: that the inter-scale forces which *generate and maintain* this stability are somehow self-evident. An atom is stable because the electromagnetic force between nucleus and electrons has a certain strength and because quantum mechanics stabilises certain orbitals. That is a precise balance between forces at different scales — nuclear forces at the femtometre scale, electromagnetic at the ångström scale. This balance is not trivial. It is the result of exactly that inter-scale coupling which the SM does not explain. The SM *uses* the stability that emerges from this coupling without grounding it.

## The planetary orbit argument

In celestial mechanics it is taken for granted that a planetary orbit is *never* a pure two-body problem. Jupiter perturbs Saturn. The Moon perturbs Earth's orbit. Tides couple rotation and orbital evolution. Small couplings accumulate over millions of years into structure-forming effects — orbital resonances, the Laplace resonance of Jupiter's moons, the stabilisation of Earth's axial tilt by the Moon. No one in celestial mechanics says: "We neglect Jupiter because it is far away." Astronomy has learned that even small couplings over long timescales are *structure-constitutive*.

At the molecular and atomic level the same logic is not applied. There, scale separation is the working principle — QCD for quarks, QED for atoms, solid-state physics for crystals, with carefully drawn boundaries between these theories. The question of whether small residual couplings between these scales become structurally visible over sufficiently long timescales or in sufficiently precise measurements is never even raised. Not because the answer is

“No”, but because the effective theories have worked so well that no one has asked what they silently integrate away.

This is not a physical necessity but a historically grown convention. Celestial mechanics and particle physics treat inter-scale couplings asymmetrically — not from a principle, but from different traditions. FFGFT claims to unify both under the same geometric logic: couplings exist at all scales, their strength scales geometrically with the logarithmic distance, and the stability of superordinate structures is not self-evident but *stands in need of explanation*.

## **Why astronomers see couplings that particle physicists do not**

There is a simple reason why inter-scale couplings are so conspicuous in celestial mechanics and so invisible in particle physics — and it has nothing to do with whether the couplings exist in principle. It has to do with *how far apart* the scales involved actually are.

In celestial mechanics, Jupiter and Saturn play on the same stage. They are both planets, both in the solar system, both on comparable orbits. The distance between their scales is tiny — almost zero when you consider the full span of nature’s scales. That is why their mutual influence is large enough to leave visible traces over millions of years.

In particle physics the scales between which one might look for couplings are unimaginably far apart. Between a quark and an atom fit eight powers of ten. Between the Planck scale and the scale of the Higgs particle fit seventeen. A coupling that acts across so many orders of magnitude grows weaker with every additional order — so weak that no instrument we have today could detect it.

This is the real reason why standard physics gets along so well with separated scales: not because the couplings are absent, but because the distances are so vast that the couplings practically

vanish. Astronomy and particle physics do not contradict each other — they are looking at different sections of the same ladder.

## **Simplicity at the bottom — why fundamental constituents are so stable**

There is a second reason why the picture becomes ever more stable going downward. It concerns not the distances between scales but the nature of what sits at each level.

The deeper one descends on the scale ladder, the *simpler* the structures become. An electron has no inner parts, no substructure, no degrees of freedom into which it could rearrange itself. A quark is similarly simple. This simplicity is not accidental — it is the reason for their extraordinary stability. A simple structure cannot decay because there is nothing simpler for it to decay into. The lowest state at a given level is stable for the straightforward reason that there is nothing below it.

The higher one climbs on the ladder, the more complex the structures become. An atom already has many degrees of freedom — electrons on different orbitals, nuclear spin, vibrational modes. A molecule has more. A protein can fold into thousands of different conformations. At the planetary scale the complexity is so enormous that any underlying geometric structure is buried under the noise of billions of atoms and their interactions.

This growth of complexity upward explains why stability becomes relative: an atom is stable under ordinary conditions but can be ionised with enough energy. A molecule is stable at room temperature but can be broken apart by heat or light. Stability diminishes the further one moves from the simple building blocks at the bottom.

## **The inversion: how much effort it takes to break simple structures**

The same situation becomes particularly tangible when viewed from the other direction.

To detach an electron from a hydrogen atom, ordinary ultraviolet light suffices — the kind of energy a lamp can produce. To split an atomic nucleus requires neutron bombardment or extreme temperatures: millions of degrees, such as those that prevail in stars or in a hydrogen bomb. That is a million times more energy. To break a proton into its quarks would require one of the largest particle accelerators in the world — and even then the quarks are never seen freely, because they immediately form new bonds.

This staggering of effort is not accidental. It shows directly how deep a structure sits on the scale ladder. The deeper, the simpler — and the more energy is needed to break it apart, precisely because there is nothing it could “fall” into. The energy threshold is, in a sense, the experimental fingerprint of depth.

Read in reverse: the fact that nuclear fission demands a million times more effort than ionising an atom is direct proof that atomic nuclei sit on a deeper, simpler level than electrons. And that even this is not the bottom level is shown by the further effort needed to isolate quarks. The scale ladder is not a metaphor — it is physically inscribed in these energy thresholds.

Standard physics describes these thresholds precisely and correctly. What it does not explain is why the spacings between the scales are exactly as large as they are — why not half as large or twice as large. FFGFT claims that these spacings follow a geometric order rather than being accidental. Whether that is true is open. That the question is legitimate, the staggering itself makes plain.

## The hierarchy problem as symptom

The most familiar symptom of this gap is the *hierarchy problem*: why is the gravitational scale (Planck mass) so many orders of magnitude larger than the electroweak scale (Higgs mass)? The SM has no explanation — it adjusts the fine-tuning empirically to the observed value. Supersymmetry, technicolour, Randall–Sundrum models: all are attempts to explain this scale gap mechanically. None has been experimentally confirmed so far.

The question this document asks — how do the scales of matter couple structurally to one another? — is therefore not only a FFGFT-specific gap. It is a *generally unsolved question in theoretical physics*. FFGFT names a mechanism intended to address it: the continuous  $\xi$ -geometry on the 4D-torus, which carries the same basic structure at every scale, with scale-specific corrections via  $K_{\text{frak}}$ . Whether this mechanism provides the answer is open. That the question is open is not.

## 2 The two reading perspectives

If one wants to describe the world, there are two possible entry perspectives. Both are old, both have a tradition in physics, and both work in their own way.

### Perspective 1 — the building-block view

The classical reductionist reading: the world consists of quarks; quarks build nucleons; nucleons build atomic nuclei; nuclei and electrons build atoms; atoms build molecules; molecules build crystals or gases or liquids; these build larger structures — minerals, rocks, planetary shells, atmospheres, hydrospheres — and finally planets, stars, galaxies, galaxy clusters, the universe as a whole. Each rung is a collection of the previous one. The bind-

ing forces between the building blocks are described locally: the strong interaction at the nuclear scale, the electromagnetic at the atomic and molecular, van-der-Waals and hydrogen bonds at the chemical-structural, gravity at the astronomical.

Standard physics moves largely in this perspective. It describes the mechanisms *within* each rung very well. It also describes the *transitions* between the rungs — solid-state physics for how atoms become crystals, hydrodynamics for how molecules become liquids, astrophysics for how matter becomes stars. What it does not describe is whether and how a geometry on one rung acts *through* the other rungs.

## Perspective 2 — the field view

The holistic reading, central in FFGFT: all rungs are *standing waves* of the same underlying vacuum field. What appears as a “building block” on one rung is, at a deeper level, a stabilized resonance configuration — an electron is a mode on the 4D-torus, an atom is a composite mode of electron and nuclear resonances, a crystal is a periodic mode over atomic modes, a planet is a gravitationally stabilized resonance of crystal aggregates. The same geometry, the same winding logic, the same mode structure holds at every rung — only at a different scale.

The FFGFT claim here is not that the building-block perspective would be wrong. It is not wrong. But it is a *local language* that has breaks between rungs which, in the field view, are no breaks. The field view has a different strength: it shows why the rungs lie *just* as they lie — not as a random accumulation of building blocks, but as a self-consistent resonance chain.

## Are both perspectives equivalent?

Both describe the same world. They are not competitors, but two languages of description for the same thing. Which is “more correct” is a misguided question — as little as a piano piece is more correctly described by a score or by an acoustic frequency spectrum. Both are exact; but they see different different things.

## 3 The scale ladder

If one orders the rungs — from smallest to largest —, there emerges a logarithmic ladder whose rungs are roughly evenly spaced. Each rung has a characteristic length, a characteristic binding energy, and a logarithmic distance to the Planck scale.

| Layer                           | char. length $\ell$ | $\log_{10}(\ell/\ell_P)$ |
|---------------------------------|---------------------|--------------------------|
| Planck scale                    | $10^{-35}$ m        | 0                        |
| Quark / lepton                  | $10^{-18}$ m        | $\sim 17$                |
| Atomic nucleus                  | $10^{-15}$ m        | $\sim 20$                |
| Atom                            | $10^{-10}$ m        | $\sim 25$                |
| Molecule                        | $10^{-9}$ m         | $\sim 26$                |
| Crystal (unit cell)             | $10^{-9}$ m         | $\sim 26$                |
| Microstructure (grain boundary) | $10^{-6}$ m         | $\sim 29$                |
| Macroscopic matter              | $10^{-3}$ m         | $\sim 32$                |
| Planetary body                  | $10^7$ m            | $\sim 42$                |
| Star                            | $10^9$ m            | $\sim 44$                |
| Solar system                    | $10^{13}$ m         | $\sim 48$                |
| Galaxy                          | $10^{21}$ m         | $\sim 56$                |
| Galaxy cluster                  | $10^{23}$ m         | $\sim 58$                |
| Universe (observable)           | $10^{26}$ m         | $\sim 61$                |

The rungs are not uniform — but they span a logarithmic range of about *61 orders of magnitude*. That is the full span over which the FFGFT geometry would have to act, if the field view is to hold throughout.

## What standard physics describes

At each rung there is an established description:

- **Quark/lepton:** Standard Model of particle physics (QCD, electroweak).
- **Atomic nucleus:** nuclear physics (shell model, mean-field, ab initio for light nuclei).
- **Atom:** quantum mechanics with Coulomb interaction; Hartree-Fock, DFT.
- **Molecule:** quantum chemistry; ab-initio methods up to about 100 atoms.
- **Crystal/solid:** solid-state physics; band model, Bloch theorem, collective excitations (phonons, magnons).
- **Liquid/gas:** statistical mechanics, hydrodynamics, thermodynamics.
- **Planetary body:** geophysics, mineralogy, classical mechanics.
- **Star/galaxy:** astrophysics, gravitational physics, cosmology.

Each of these theories is very accurate at its scale. But each works with its *own vocabulary*, its own constants, its own interactions. Standard physics gives no continuous language connecting all scales — it welds them together by transition theories.

## What FFGFT would add

FFGFT would say: at *every* of these rungs the same vacuum parameter  $\xi_0 = 4/30000$  acts. It acts everywhere with the same local strength (this is the statement of Layer 1, Doc. 241). But the *cumulative* effect over the scale flow between the rungs is different, depending on the logarithmic distance. This is the natural

generalization of what Doc. 133 makes explicit for the electroweak scale:

$$K_{\text{frak}}(\Lambda_1 \rightarrow \Lambda_2) = 1 - \delta \cdot \xi \cdot \ln(\Lambda_1/\Lambda_2),$$

with  $\delta$  a scale-specific coefficient that depends on the geometric averaging over the rungs. For the jump Planck  $\rightarrow$  electroweak this cumulated correction is about 1.3%. For jumps at higher scales the corrections are *smaller*, not larger — see the next section.

## 4 The central hypothesis: small but continuous couplings

Here comes the central point of this document, and it is honestly flagged as **hypothesis, not proof**.

### Statement

The FFGFT geometry acts on all scales, but the direct geometric couplings *between* scales far apart from one another are *very small*. This is a necessary consequence of the architecture:

1. The parameter  $\xi$  itself is small ( $\sim 10^{-4}$ ).
2. Coupling between scales enters only via the logarithmic distance — not the linear distance.
3. Even within the 17-order-of-magnitude jump (Planck  $\rightarrow$  electroweak), the cumulated correction is only 1.3%.
4. Between scales only a few orders apart (e.g., atom  $\rightarrow$  crystal:  $\sim 1$  order of magnitude), the direct geometric coupling is smaller by factors of  $\sim 100$  — that is,  $\sim 0.01\%$  or below.

Concretely this means: a geometry at the atomic scale “shines through” into the macroscopic world only as a correction at the order of  $10^{-4}$  to  $10^{-6}$  relative to the dominant interaction of the

respective scale. Standard physics mostly does not “see” these corrections, because they sit below its resolution and are swept away by local renormalization or averaging procedures.

## Why this justifies the building-block perspective

Precisely because the inter-scale couplings are small, the building-block perspective of standard physics is *locally very accurate*. Whoever does atomic physics can ignore that the same geometry also acts in a galaxy cluster; the influence lies beyond any spectroscopic resolution. Whoever does crystallography can ignore that the same vacuum parameter also appears in electroweak theory. The scale separation of standard physics is not wrong; it is a very good approximation.

FFGFT does not contradict the scale separation. It only adds a further statement: the separation is *not absolute*. At sufficiently high precision, traces of the continuous geometry should appear — as systematic deviations that cannot be explained by local mechanisms.

## Where such traces might be observable

The following areas are the most plausible candidates where inter-scale FFGFT couplings could become experimentally visible, if the hypothesis is correct:

- High-precision spectroscopy of atoms and molecules (anomalous shifts beyond the QED corrections, order  $10^{-6}$  relative).
- Anomalous corrections in solid-state band structures that cannot be explained by local many-body effects.

- Correlations between macroscopic constants (e.g., material densities, critical points, phase-transition energies) and atomic quantities, beyond known scaling laws.
- Astronomical ratios (planetary spacings, stellar mass-radius relations) that systematically differ slightly from purely classical-gravitational predictions.

All four points are formulated as research directions, not as predictions. No one has so far systematically searched for such couplings in the magnitude range  $10^{-4}$  to  $10^{-6}$ , because no established theory expects them.

## 5 The multi-storey concert hall picture

Whoever knows the concert-hall picture from Doc. 159 can see here a structural extension: *the concert hall has multiple storeys*.

On each storey there is an acoustic room of its own, with its own resonances, its own reverberation time, its own characteristic frequency. A tone played on the ground floor has its dominant resonances on the ground floor. But it penetrates faintly, through walls and ceilings, into the first storey and the second. What is a full sound on the ground floor is in the second storey only a weak sympathetic vibration — but it is not zero. And conversely: a loud tone on the top floor penetrates faintly downwards.

That is how to picture the scales: atomic geometry has its resonances at the atomic rung, solid-state geometry at the solid-state rung, planetary geometry at the planetary rung. But each rung vibrates faintly in the others — because they all stand on the same vacuum, in the same fractal geometry. The direct couplings are small because between widely separated rungs many “walls and ceilings” damp them. But they are not zero.

In this language, standard physics is the theory that describes each storey *by itself* very well. FFGFT is the theory that asks how the building sounds as a whole — and that claims that there are quiet cross-couplings which a single storey cannot explain.

## 6 The two languages and their synthesis

This allows the question stated at the outset — how do building-block view and field view connect without metaphor? — to be answered in the following form:

- The *building-block view* describes each rung *locally* and its direct transitions to the next-higher or next-lower rung. It is the language of local mechanisms.
- The *field view* describes the common vacuum on which all rungs stand, and the continuous geometry that is the same on all rungs. It is the language of global consistency.
- Both are exact at their own resolution level. The building-block view is more accurate at describing a single rung. The field view is more accurate at the question of why the rungs lie as they lie, and how they are self-consistently connected.
- The direct inter-scale couplings are *small* — so small that the building-block view practically always suffices. But they are not zero — and at sufficiently high experimental precision they should become visible as systematic corrections.

## 7 What we do not say

We do *not* say:

- that standard physics is wrong;

- that all mechanisms between scales can be derived from FFGFT alone;
- that the multi-storey concert-hall picture is a complete theory;
- that the coupling strengths are quantitatively known from FFGFT — they are *not* derivable from the current documents, only plausible as orders of magnitude.

We do say, honestly as hypothesis:

- that the geometry of the vacuum acts on all scales;
- that the scale separation of standard physics is a very good but *not perfect* approximation;
- that the inter-scale couplings are small but in principle searchable in high-precision experiments;
- that this aspect of FFGFT is open research terrain.

## 8 What would need to be done

To turn this sketch into a testable theory, three steps would be necessary, to be taken in future documents:

1. *A quantitative form of the coupling formula* — how does the cumulated correction  $K_{\text{frak}}^{(n,m)}$  between two scales  $\Lambda_n$  and  $\Lambda_m$  depend explicitly on the logarithmic distance? Doc. 133 contains the case Planck  $\rightarrow$  electroweak; the generalization is outstanding.
2. *A selection of concrete observation sites* where the inter-scale coupling would be best testable, and a prediction of the expected effect size.
3. *A comparison with standard physics* in which it is honestly marked where FFGFT makes a prediction that goes beyond

the Standard Model — and where it merely provides an alternative language for the same fact.

All three steps are non-trivial. They are what lies between a sketch such as this document and a fully developed theory. This document carries out none of these steps; it only marks them as necessary.

## Summary

- FFGFT is so far worked out at the lowest layer (elementary particles) and at the highest (cosmology); the intermediate layers — atoms, molecules, crystals, matter aggregates, planets, stars, galaxies — are only addressed in scattered places.
- The world can be read in two complementary languages: building-block view (local, standard physics) and field view (global, FFGFT). Both are exact; they see different different things.
- The FFGFT geometry, by assumption, acts on all scales with the same local strength. The cumulative effect over scale distances is small because the distances are large: traces of order  $10^{-4}$  to  $10^{-6}$  relative.
- This justifies the scale separation of standard physics as a very good approximation — but not as a principled boundary.
- Status: **hypothesis, not proof**. Quantitative generalization of  $K_{\text{frak}}$  to arbitrary scale pairs and concrete predictions are outstanding.

*Status: May 2026. This document is methodologically pedagogical and explicitly marked as a plausibility sketch. It contains no independent quantitative predictions.*